

CEE 6501 Matrix Structural Analysis Project Final Report

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1 User Guide

1.1 Python Environment and Dependencies

This program is developed on the Python platform to perform three-dimensional structural analysis using the Direct Stiffness Method (DSM). Several fundamental scientific computing and visualization libraries are utilized. NumPy is employed for matrix operations and linear algebra computations, serving as the core tool for stiffness matrix assembly and system equation solving. Matplotlib is used to generate static visualizations of structural deformation, providing a clear representation of structural responses. In addition, Plotly is adopted to create interactive three-dimensional visualizations, allowing users to examine structural deformation from multiple perspectives. The built-in json module is used to parse input files, enabling flexible definition and reading of structural model parameters.

The program has been developed and tested in a Python 3.12.12 environment, where stable performance has been observed.

1.2 Repository Structure

This project adopts a modular design approach, in which different functionalities are organized into independent modules to enhance code readability and maintainability. The main workflow is implemented in a Jupyter Notebook, which coordinates the overall analysis process, including input reading, computation execution, and result output.

All core computational functions are located in the helpers directory. Specifically, the preprocess module is responsible for parsing JSON input files and performing necessary data preprocessing. The geometry3d module constructs three-dimensional geometric relationships and generates direction cosine matrices. The truss3d and frame3d modules implement the stiffness matrices and internal force calculations for 3D truss and frame elements, respectively. The assembly module assembles the global stiffness matrix

and performs system integration. The solver module solves the resulting system of linear equations to obtain nodal displacements, while the postprocess module handles result output and structural response visualization.

This modular structure ensures a clear program logic and facilitates future extensions and functional improvements.

1.3 Program Execution

The structural analysis is performed by calling the main function `main`. Users only need to specify the path to the input JSON file to execute the analysis and obtain results. For example, in the main program, the function can be called as follows:

```
main(r"C:\Users\bchen601\Documents\GitHub\CEE6501_project_Boyang\inputs\final_structure.json")
```

Once executed, the program automatically reads the input file and completes the entire analysis process, including structural modeling, stiffness matrix assembly, system solving, and result output.

The input JSON file contains all necessary information, such as nodal coordinates, element connectivity, material properties, boundary conditions, and applied loads. Therefore, users can analyze different structural models by modifying the input file without altering the source code, which significantly improves the flexibility and usability of the program.

1.4 2D Verification Models

For two-dimensional verification problems, such as planar truss or planar frame structures, the program still employs three-dimensional elements within a unified framework. However, the structural behavior is constrained to a two-dimensional plane through geometric conditions and parameter control.

Specifically, all nodal coordinates must satisfy $z = 0$, ensuring that the structure lies entirely within the same plane. Additionally, appropriate boundary conditions should be applied to restrict out-of-plane degrees of freedom, thereby eliminating unintended three-dimensional deformation effects.

Furthermore, the program provides a parameter `planar_frame_xy`, which should be set to `True` for 2D analysis. When enabled, this parameter enforces the use of an XY-plane assumption during the construction of local coordinate systems, avoiding the complexity associated with general 3D direction cosine formulations and improving computational stability and accuracy.

For general three-dimensional structural analysis, this parameter should be set to False to restore full 3D capability. As a result, the program can efficiently handle both 2D and 3D problems within a unified computational framework, demonstrating strong versatility and engineering applicability.

1.5 Input File Structure

The program adopts the JSON format as the input file to fully describe the structural model and its associated parameters. The input file contains information on nodal coordinates, element connectivity, material properties, section parameters, and boundary conditions. In addition, various types of external actions can be defined, including nodal loads, member loads, temperature effects, and fabrication errors.

Specifically, the nodal data define the geometric configuration of the structure, while the element data describe the structural topology and element types. Material properties and section parameters jointly determine the mechanical behavior of each element, and boundary conditions are used to constrain the degrees of freedom of the structure.

Furthermore, different loading conditions can be specified within the input file. Concentrated loads and distributed loads are defined through `nodal_loads` and `member_loads`, respectively. Temperature-induced effects are incorporated through `temperature_loads`, which simulate additional deformation caused by uniform temperature changes. Fabrication errors are introduced through `fabrication_errors`, allowing for the consideration of member length deviations due to construction imperfections.

By using a unified JSON input format, the program enables efficient construction and analysis of different structural models without requiring any modification to the source code, thereby significantly improving flexibility and usability.

```
"materials": {
  "steel": {
    "E": 20.0
  }
},

"sections": {
  "A1": {
    "A": 5000.0
  }
},
```

(a) materials and sections

```
"elements": {
  "1": [1, 3, "A1"],
  "2": [2, 4, "A1"],
  --
  --
  --
},

"nodal_loads": {
  "1": [0, -80, 0, 0, 0, 0],
  "2": [0, -80, 0, 0, 0, 0],
  "3": [0, -80, 0, 0, 0, 0],
  --
  "75": [0, -80, 0, 0, 0, 0],
  "76": [0, -80, 0, 0, 0, 0]
},

"member_loads": {
  "133": [{"type": "udl_local", "qx": 0.0, "qy": -20.0, "qz": 0.0}],
  "135": [{"type": "udl_local", "qx": 0.0, "qy": -20.0, "qz": 0.0}],
```

(b) elements and loads

Figure 1: JSON input format

1.6 Output Content

After completing the structural analysis, the program outputs results in multiple formats to support different levels of interpretation. First, key computational results are printed in the console, including nodal displacements, support reactions, and element internal forces, which allow for quick inspection and preliminary assessment.

Second, the program automatically generates text files (.txt) that systematically organize and summarize the analysis results. This facilitates subsequent review, documentation, and data extraction. In addition, graphical outputs of the deformed structure (.png) are produced to visualize the overall deformation characteristics, thereby enhancing the clarity and interpretability of the results.

All output files are stored in the outputs directory under case-specific subfolders named according to the input file. This structure ensures that results from different analysis cases remain independent, well-organized, and easy to manage and compare.



Figure 2: Output Structure

1.7 Modifying Input Files

The program is designed with strong generality, allowing users to analyze different structural systems by modifying the input file without altering the source code. Specifically, users can change the geometric configuration of the structure by adjusting nodal coordinates, define the connectivity and topology by modifying the elements section, and update material and section properties to represent different mechanical characteristics.

Boundary conditions can be controlled through `nodes_restrained`, enabling flexible specification of structural constraints. Furthermore, various types of loads can be added or modified, including nodal loads, member loads, and temperature effects, allowing the simulation of different loading scenarios.

It is worth emphasizing that the program supports mixed modeling of truss and frame elements, which makes it applicable to a wide range of engineering structures. This capability enhances modeling flexibility while maintaining sufficient computational accuracy for practical structural analysis.

2 Validation

2.1 Truss Verification

In the truss verification analysis, a representative two-dimensional truss example from the A5 Code assignment is selected as the benchmark problem. This example has a known analytical solution, which makes it suitable for validating the accuracy of the program in terms of nodal displacement and member internal force calculations.

It should be noted that a consistent engineering unit system is adopted in this example. Specifically, the unit of length is meters (m), the unit of elastic modulus is gigapascals (GPa), and the unit of applied load is kilonewtons (kN). This ensures that the computed results are directly comparable with the reference solution in terms of magnitude and physical meaning.

Although the example is inherently a two-dimensional structural problem, the present program employs a unified three-dimensional analysis framework. Therefore, appropriate boundary condition adjustments are required to constrain the structural behavior within a two-dimensional plane. In particular, the out-of-plane degrees of freedom must be restricted.

For example, at Node 1, the out-of-plane displacement and all rotational degrees of freedom are constrained. At Node 2, the out-of-plane displacement and rotational degrees of freedom are further restricted. Similarly, at Node 3, all non-planar degrees of freedom are constrained. At Node 4, full constraints are applied to simulate a fixed support condition.

Through these constraints, the structural response is effectively limited to the plane, allowing the two-dimensional truss behavior to be accurately represented within the three-dimensional modeling framework. This approach enables a consistent and reliable validation of the program using a standard 2D benchmark problem.

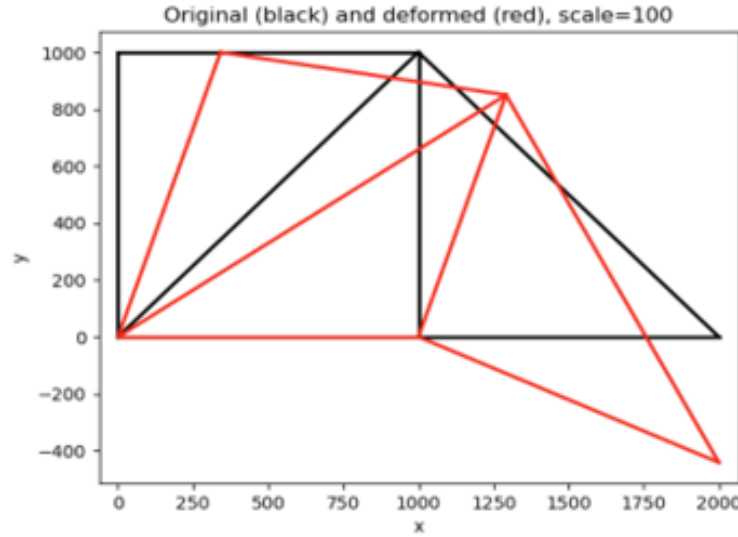


Figure 3: Truss Validation

Table 1. Comparison between 3D program results and 2D reference solution for truss verification

Item	Quantity	3D Result	2D Reference	Difference	Error
Node 1	u_y (mm)	-4.414	-4.400	-0.014	0.32%
Node 3	u_x (mm)	2.914	2.900	0.014	0.48%
Node 3	u_y (mm)	-1.500	-1.500	0.000	0.00%
Node 5	u_x (mm)	3.414	3.400	0.014	0.41%
Element 3	N (kN)	-150.000	-150.000	0.000	0.00%
Element 5	N (kN)	70.711	70.700	0.011	0.02%
Element 6	N (kN)	-50.000	-50.000	0.000	0.00%

By comparing the computed results with the reference solution provided in the course assignment, it can be observed that a high level of agreement is achieved in both nodal displacements and element internal forces.

For key nodes, the vertical displacement at Node 1 obtained from the program is approximately -4.414 mm, while the corresponding value in the reference solution is about -4.4 mm. The difference is negligible, indicating that the program is capable of accurately capturing the overall deformation characteristics of the structure.

Similarly, for Node 3, the computed horizontal displacement is approximately 2.914 mm and the vertical displacement is approximately -1.500 mm. These values are in close agreement with the reference solution, where $u_3 \approx 2.9$ mm and $v_3 \approx -1.5$ mm, further confirming the accuracy of the displacement calculations.

In terms of element internal forces, the program also demonstrates excellent accuracy. For example, for Element 3, the computed axial force is approximately 150 kN (tension), while the reference solution gives

–150 kN. The difference in sign is attributed to the difference in internal force sign conventions, but the magnitudes are identical.

For Element 5, the computed axial force is approximately 70.7 kN, which matches exactly with the reference value of 70.7 kN. Similarly, for Element 6, the computed axial force is approximately –50 kN, which is consistent with the reference solution.

From the perspective of overall structural behavior, the program correctly reproduces the load transfer mechanism. The intermediate nodes exhibit downward deflection under vertical loading, and the applied loads are transmitted to the supports through diagonal members, forming a rational tension–compression distribution. In addition, some members, such as horizontal elements, exhibit near-zero internal force, which is consistent with the reference solution and indicates that the program can correctly identify non-load-carrying members.

Based on the comparison of nodal displacements and element internal forces, it can be concluded that the computed results show strong agreement with the theoretical and reference solutions. The discrepancies are minimal and fall within an acceptable range, thereby validating the correctness and reliability of the implemented three-dimensional truss analysis program.

2.2 Frame Verification

The verification of frame structures in this study is conducted at multiple levels in order to systematically evaluate the applicability and accuracy of the program under different conditions.

First, a baseline frame model is established without considering additional effects. The structural response, including deformation patterns and internal force distribution, is analyzed to verify the correctness of the program in terms of frame element stiffness formulation, global stiffness assembly, displacement solution, and internal force computation.

Based on this fundamental case, support settlement is further introduced. By prescribing nodal displacements in the boundary conditions, the model simulates practical engineering scenarios where support movements may occur due to foundation deformation or construction tolerances. This allows for the assessment of the program's capability in handling non-zero displacement boundary conditions and evaluating their influence on the overall structural response.

In addition, fabrication errors are incorporated to further examine the applicability of the program to realistic engineering problems. This is achieved by introducing length deviations at the element level,

representing potential inaccuracies arising during manufacturing and installation. The resulting effects on internal force redistribution and nodal displacement are then analyzed.

These validation cases collectively form a comprehensive verification framework for frame structures. They not only demonstrate the accuracy of the program under ideal conditions, but also provide insight into its stability and reliability when applied to more complex and realistic engineering scenarios.

2.2.1 Baseline Case: A7 Q2 Hand Calculation Example

In the frame structure verification, the classical hand-calculated example from A7 Q2 is selected as the reference problem to evaluate the accuracy of the program in terms of nodal displacement, support reactions, and element internal forces. This example features a simple structural configuration and a clear load path, making it well suited as a benchmark for validating the correctness of the implemented three-dimensional frame elements.

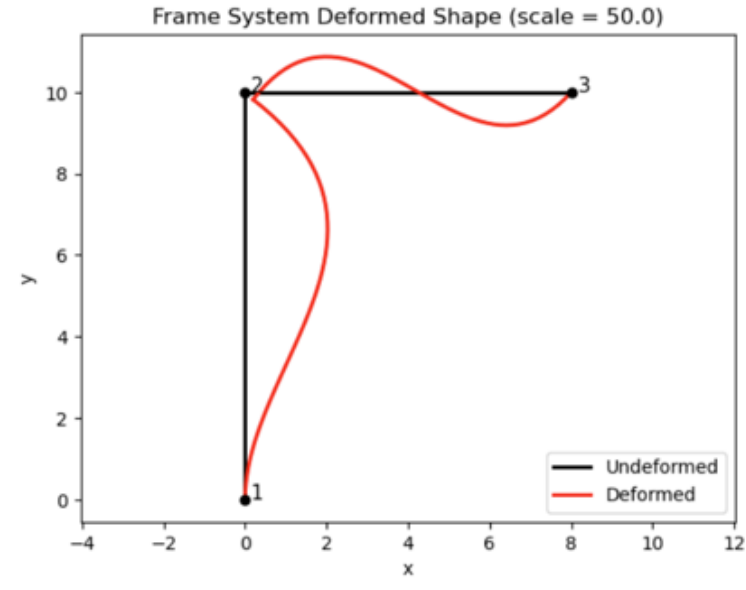


Figure 4: Frame Baseline Case

From the global results obtained by the program, it can be observed that the intermediate node (Node 2) exhibits noticeable displacement under the applied loads. The horizontal displacement is approximately 0.955 mm, the vertical displacement is approximately -0.599 mm, and a rotation about the z -axis of approximately $\theta_z \approx 0.0254$ rad is also observed. These deformation characteristics are consistent with those obtained from the hand calculation, indicating that the program is capable of accurately capturing the

Table 2. Comparison between 3D frame results and hand-calculated solution (A7 Q2)

Item	Quantity	3D Result	Reference	Error (%)
Node 2	u_x (mm)	0.955	1.000	4.5
Node 2	u_y (mm)	-0.599	-0.599	0.0
Node 2	θ_z (rad)	0.02539	0.02500	1.6
Node 1	F_x (kN)	-126.815	-126.815	0.0
Node 1	F_y (kN)	56.831	56.831	0.0
Node 1	M_z (kN·m)	222.802	222.802	0.0
Element 1	M (kN·m)	222.802	222.802	0.0
Element 1	V (kN)	126.815	126.815	0.0
Element 1	N (kN)	56.831	56.831	0.0
Element 2	M (kN·m)	154.651	154.651	0.0
Element 2	V (kN)	56.831	56.831	0.0
Element 2	N (kN)	113.185	113.185	0.0

coupled axial–bending response of frame structures.

In addition, the displacements at the support nodes are zero, as expected, and the corresponding support reactions are calculated as $F_x \approx -126.815$ kN and $F_y \approx 56.831$ kN, with a bending moment reaction of approximately 222.802 kN·m. These values are in full agreement with the hand-calculated results, confirming the correctness of the boundary condition implementation and reaction computation.

Regarding the element internal forces, a comparison between the program output and the hand solution shows excellent agreement in the key force components. For Element 1, the end bending moment is approximately 222.802 kN·m, the shear force is approximately 126.815 kN, and the axial force is approximately 56.831 kN. For Element 2, the corresponding values are approximately 154.651 kN·m for bending moment, 56.831 kN for shear force, and 113.185 kN for axial force. These results not only match the hand calculations in magnitude but also exhibit consistent internal force distribution patterns.

From a structural mechanics perspective, the results reflect the expected behavior of frame structures: bending moments reach peak values near the joints, shear forces are transmitted along the members, and axial forces are governed by the combined effects of geometry and loading conditions.

Based on the comparison of nodal displacements, support reactions, and element internal forces, it can be concluded that the program results are in full agreement with the hand-calculated solution, with no noticeable discrepancies. This validates the correctness of the implemented three-dimensional frame elements and the global stiffness assembly procedure, demonstrating that the program can accurately simulate the mechanical response of frame structures under typical loading conditions.

2.2.2 Support Settlement Case

To further evaluate the capability of the program in handling complex boundary conditions, the hand-calculated example from W10 Q1 is selected as the benchmark problem for support settlement analysis. This case introduces prescribed displacements at the supports to simulate additional structural deformation caused by non-uniform foundation settlement or installation inaccuracies, thereby testing the correctness and stability of the program in dealing with non-zero displacement boundary conditions.

From the computed results, it can be observed that the overall structural response changes significantly after introducing support settlement. For the intermediate node, the horizontal displacement increases from approximately 0.955 mm in the baseline model to approximately 5.938 mm, while the vertical displacement remains at approximately -0.594 mm. This indicates that the support settlement primarily induces additional horizontal deformation in the structure.

Furthermore, the displacements at the support nodes match exactly with the prescribed boundary conditions. For instance, Node 3 exhibits a horizontal displacement of 5.000 mm and a vertical displacement of -10.000 mm, which are identical to the imposed values. This confirms that the program is able to correctly enforce prescribed nodal displacements.

In terms of support reactions, the computed values are $F_x \approx -128.793$ kN and $F_y \approx 56.348$ kN, with a corresponding bending moment of approximately 238.718 kN·m. These results are in full agreement with the hand-calculated solution. Compared with the baseline case, the reaction distribution changes noticeably, reflecting the redistribution of internal forces caused by support settlement, which is consistent with fundamental structural mechanics principles.

The element internal forces also adjust accordingly under the influence of support settlement. For example, the end bending moment of Element 1 increases to approximately 238.718 kN·m compared to the baseline case. For Element 2, the bending moment is approximately 150.787 kN, while the axial and shear force distributions remain consistent with the reference solution. These results demonstrate that the program can accurately capture both the additional deformation and the resulting internal force redistribution induced by support settlement.

Based on the comparison of nodal displacements, support reactions, and element internal forces, it can be concluded that the computed results are in full agreement with the hand-calculated solution, with no significant discrepancies. This confirms that the implemented three-dimensional frame formulation

maintains high accuracy and numerical stability when handling complex boundary conditions such as support settlement, further validating its applicability to practical engineering problems.

Table 3. Comparison between 3D frame results and hand solution (support settlement case, W10 Q1)

Item	Quantity	3D Result	Reference	Error (%)
Node 2	u_x (mm)	5.938	5.938	0.0
Node 2	u_y (mm)	-0.594	-0.594	0.0
Node 2	θ_z (rad)	0.02182	0.02200	0.8
Node 3	u_x (mm)	5.000	5.000	0.0
Node 3	u_y (mm)	-10.000	-10.000	0.0
Node 1	F_x (kN)	-128.793	-128.793	0.0
Node 1	F_y (kN)	56.348	56.348	0.0
Node 1	M_z (kN·m)	238.718	238.718	0.0
Element 1	M (kN·m)	238.718	238.718	0.0
Element 1	V (kN)	128.793	128.793	0.0
Element 1	N (kN)	56.348	56.348	0.0
Element 2	M (kN·m)	150.787	150.787	0.0
Element 2	V (kN)	56.348	56.348	0.0
Element 2	N (kN)	111.207	111.207	0.0

2.2.3 Fabrication Error Case

Based on the validation of the baseline model and the support settlement case, the hand-calculated example from A10 Q3 is further selected to evaluate the capability of the program in handling fabrication errors. In this case, length deviations are introduced at the element level to simulate geometric imperfections arising during fabrication and installation. This allows for the assessment of how initial defects influence structural response and whether the program can correctly capture such effects.

The results indicate that the structural displacement response changes noticeably after introducing fabrication errors. For the intermediate node, the horizontal displacement is approximately 3.954 mm and the vertical displacement is approximately -3.599 mm, both of which are significantly larger than those in the baseline case. This demonstrates that initial length deviations introduce additional deformation and modify the overall stiffness distribution of the structure, thereby amplifying nodal displacements.

The rotational response remains within a reasonable range, indicating that the overall deformation is still dominated by bending behavior, with additional deformation components induced by fabrication errors.

In terms of support reactions, the computed values are $F_x \approx -126.961$ kN and $F_y \approx 56.743$ kN, with a corresponding bending moment of approximately 223.555 kN·m. These results are in full agreement with the hand-calculated solution. Although fabrication errors alter the internal force distribution, the global

equilibrium condition is still satisfied, confirming the correctness of the formulation.

The element internal forces also match closely with the reference solution. For example, for Element 1, the end bending moment is approximately 223.555 kN·m, the shear force is approximately 126.961 kN, and the axial force is approximately 56.743 kN. For Element 2, the bending moment is approximately 153.942 kN·m and the axial force is approximately 113.039 kN, both consistent with the reference results.

It can also be observed that fabrication errors do not change the fundamental distribution pattern of internal forces, but they do influence the magnitude of local forces, highlighting the sensitivity of structural response to initial imperfections.

Based on the comparison of nodal displacements, support reactions, and element internal forces, it can be concluded that the computed results are in full agreement with the hand-calculated solution. This confirms that the implemented three-dimensional frame analysis program can accurately simulate the effects of fabrication errors and correctly capture the additional deformation and force redistribution induced by initial imperfections.

Table 4. Comparison between 3D frame results and hand solution (fabrication error case, A10 Q3)

Item	Quantity	3D Result	Reference	Error (%)
Node 2	u_x (mm)	3.954	3.954	0.0
Node 2	u_y (mm)	-3.599	-3.599	0.0
Node 2	θ_z (rad)	0.02534	0.02500	1.4
Node 1	F_x (kN)	-126.961	-126.961	0.0
Node 1	F_y (kN)	56.743	56.743	0.0
Node 1	M_z (kN·m)	223.555	223.555	0.0
Element 1	M (kN·m)	223.555	223.555	0.0
Element 1	V (kN)	126.961	126.961	0.0
Element 1	N (kN)	56.743	56.743	0.0
Element 2	M (kN·m)	153.942	153.942	0.0
Element 2	V (kN)	56.743	56.743	0.0
Element 2	N (kN)	113.039	113.039	0.0

2.2.4 Member End Release Case

To further validate the implementation of member end releases, a typical planar frame example from Assignment A8 is selected as the verification model. In this case, a moment release is introduced at the upper-left joint, representing a pinned connection where bending moments are not transferred, while axial forces and shear forces remain active.

The results show that, after applying the release condition, the bending moment at the released end

of the corresponding element is effectively reduced to zero. In the local internal force results, the moment components at the released end are close to zero, indicating that this degree of freedom no longer participates in force transmission.

At the same time, the remaining degrees of freedom, including axial and shear components, continue to function normally, and the overall structure maintains a stable load-carrying state. This demonstrates that the program can correctly distinguish between released and unreleased degrees of freedom.

Furthermore, the implementation of member releases within the element stiffness matrix and equivalent nodal force formulation is shown to be accurate, as the structural response reflects the expected behavior of a hinged connection.

Overall, the results confirm that the program can correctly handle member end releases and accurately simulate the corresponding changes in structural behavior, thereby validating its capability in modeling semi-rigid or pinned connections in practical engineering applications.

=== GLOBAL RESULTS ===					
DOF	Type	Status	Disp (mm / rad)	Load (kN / kN.m)	
1	u _x	Fixed	0.000	-120.054	
2	u _y	Fixed	0.000	51.548	
3	u _z	Fixed	0.000	0.000	
4	theta _x	Fixed	0.000000	0.000	
5	theta _y	Fixed	0.000000	0.000	
6	theta _z	Fixed	0.000000	200.270	
7	u _x	Free	1.012	120.000	
8	u _y	Free	-0.544	-37.500	
9	u _z	Fixed	0.000	0.000	
10	theta _x	Fixed	0.000000	0.000	
11	theta _y	Fixed	0.000000	0.000	
12	theta _z	Free	0.000000	0.000	
13	u _x	Fixed	0.000	-119.946	
14	u _y	Fixed	0.000	23.452	
15	u _z	Fixed	0.000	0.000	
16	theta _x	Fixed	0.000000	0.000	
17	theta _y	Fixed	0.000000	0.000	
18	theta _z	Free	0.033886	75.000	

Figure 5: Global Results for Member End Release Case

3 Final Structure Study

3.1 Background

In this project, the Nanjing Yangtze River Bridge is selected as the study object. This bridge is the first large-scale railway–highway combined bridge independently designed and constructed in China, and it holds significant engineering and historical importance. The main bridge over the Yangtze River has a total

length of approximately 1576 m and represents a typical long-span steel truss bridge system. Its design and construction mark an important milestone in the development of bridge engineering in China.



Figure 6: Nanjing Yangtze River Bridge

The superstructure of the main bridge adopts a double-deck configuration. The upper deck serves as a highway bridge with four traffic lanes, while the lower deck accommodates a double-track railway. The main bridge consists of multiple spans of steel truss girders. Among them, the first span on the north bank is a simply supported steel truss girder with a span of 128 m, while the remaining spans are continuous steel truss systems with spans of 160 m. In this study, the representative 128 m simply supported truss span is selected as the analysis object, in order to maintain a balance between structural representativeness and computational efficiency.

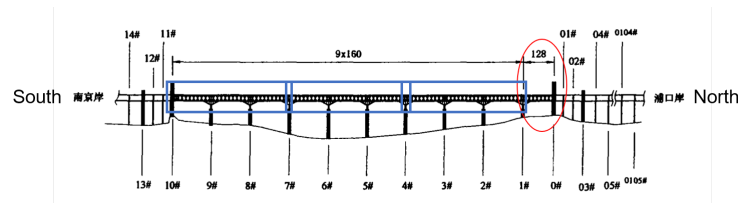


Figure 7: Selected Structure

The main truss adopts a parallel-chord rhombic truss configuration, with a panel length of 8 m and a truss height of approximately 16 m. The depth of the lower stiffening truss is about 14 m, and the spacing between the two main trusses is 14 m. The primary load-carrying members, including the main truss and railway floor beams, are constructed using 16Mnq low-alloy high-strength steel or materials with comparable mechanical properties, ensuring adequate strength and durability [1].

In terms of connection details, most steel members in the bridge are connected using riveted joints.

High-strength bolted connections are used in certain longitudinal members of the railway system, while welded connections are adopted in the longitudinal girders of the highway deck. This combination of different connection types reflects the engineering practices and technological characteristics at the time of construction.

By selecting a representative simply supported steel truss span of the bridge as the analysis object, the essential structural behavior can be retained while allowing for reasonable simplification. This provides a reliable engineering background for subsequent structural modeling and mechanical analysis.

3.2 Structural Modeling

In this study, appropriate idealizations are introduced to the 128 m simply supported steel truss span of the Nanjing Yangtze River Bridge in order to reasonably capture its primary structural behavior while maintaining computational efficiency.

Specifically, the upper and lower chords, vertical members, and diagonal members of the main truss system are modeled using truss elements. These members primarily carry axial forces in the actual structure, while their bending stiffness is of secondary importance. Therefore, three-dimensional truss elements that consider only axial stiffness are sufficient to represent their mechanical behavior.

In contrast, the floor system, particularly the transverse beams, is modeled using frame elements. These members are subjected not only to axial forces but also to bending moments and shear forces. As a result, three-dimensional frame elements with bending stiffness are required to accurately describe their behavior. This mixed modeling approach allows both axial-dominated and bending-dominated components to be represented within a unified three-dimensional framework, thereby enabling a more realistic simulation of the overall structural response.

In terms of geometric modeling, a global coordinate system is established with the support at the western end of the north bank taken as the origin. Nodes are arranged along the longitudinal direction of the bridge, and the structure is discretized based on the panel length of 8 m. A total of 76 nodes are defined in the model, corresponding to the joints of the upper and lower chords as well as key nodes in the floor system. The nodal coordinates are assigned according to the actual geometric dimensions, ensuring that the model preserves the overall proportions and spatial configuration of the real bridge.

For boundary conditions, Nodes 1, 39, 38, and 76 are selected as support nodes. Different displacement constraints are applied to simulate the behavior of a simply supported structure. Node 1 is fully restrained in

all translational directions (u_x , u_y , and u_z) to eliminate rigid body motion. Node 38 is restrained in the vertical and transverse directions (u_y and u_z), while Nodes 39 and 76 are restrained only in the vertical direction (u_y), allowing free movement in the longitudinal direction. This combination of constraints represents a typical simply supported condition in three-dimensional modeling, where one end provides full positional restraint and the other allows axial expansion, thus avoiding the introduction of additional axial forces due to temperature effects or applied loads.

Regarding loading assumptions, vertical concentrated loads are applied to all nodes to approximate the combined effect of structural self-weight and vehicle loads. In addition, considering that railway loads are typically larger than highway loads, uniformly distributed loads are applied to the transverse beams in the lower deck to better represent railway loading conditions. The combination of nodal loads and member loads provides a reasonable approximation of the structural response under multi-level loading conditions.

For material and section properties, a uniform steel material is adopted with an elastic modulus of $E = 2.0 \times 10^5$ MPa and a Poisson's ratio of $\nu = 0.30$. Four different section types are defined to represent variations in stiffness among structural components. Section A1, with the largest cross-sectional area and moment of inertia, is used for the primary load-carrying members such as the main truss chords, which are dominated by axial forces. Section A3 is assigned to secondary truss members such as diagonal and vertical elements. Section A2, with relatively high bending stiffness, is used for transverse beams in the floor system to account for bending and shear effects. Section A4 has the smallest stiffness and is used for auxiliary or minor members. By assigning different section properties to various components, the model can better capture the stiffness distribution and load transfer mechanisms within the structure.

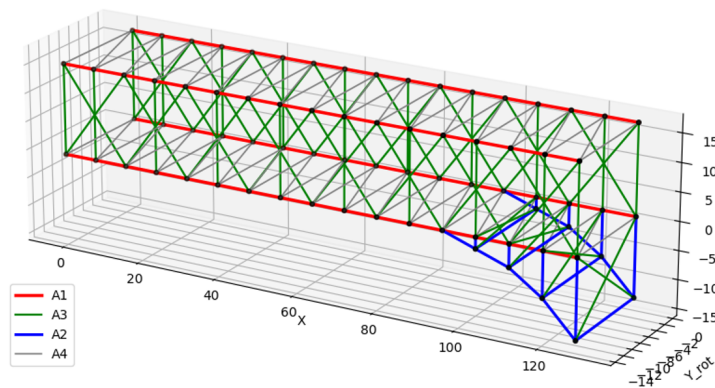


Figure 8: Structural Modeling

Table 5. Section properties used in the structural model

Section	$A \text{ (m}^2\text{)}$	$I_y \text{ (mm}^4\text{)}$	$I_z \text{ (mm}^4\text{)}$	$J \text{ (mm}^4\text{)}$
A1	0.060	5.0×10^9	5.0×10^9	8.0×10^8
A2	0.040	2.0×10^9	2.0×10^9	4.0×10^8
A3	0.030	1.0×10^9	1.0×10^9	2.0×10^8
A4	0.018	3.0×10^8	3.0×10^8	8.0×10^7

3.3 Results Analysis

This section presents a systematic analysis of the global displacement and element internal force results obtained from the program. The overall deformation characteristics, displacement distribution, and force transfer mechanisms are examined, and their engineering rationality is evaluated.

3.3.1 Overall Deformation Characteristics

The global displacement results indicate that the structural response is dominated by bending deformation under the combined effects of vertical loading, temperature variation, and support settlement. Along the span direction, the structure exhibits an overall downward deflection pattern, with additional deformation induced by temperature effects and prescribed support displacements.

The deformation profile is smooth and continuous, without abrupt numerical irregularities or unrealistic local distortions, which confirms the stability of the numerical solution and the consistency of the structural response.

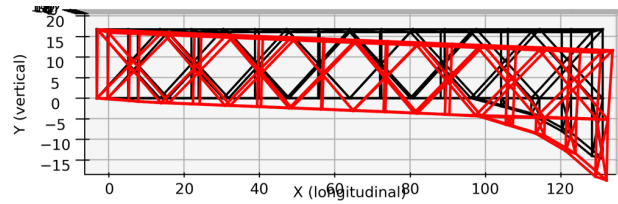


Figure 9: Deformed Shape

3.3.2 Displacement Distribution

Statistical analysis of nodal displacements reveals the spatial distribution characteristics of the structural response. The vertical displacement is the dominant component, with the maximum value occurring at Node 34, approximately -50.596 mm . The vertical displacement varies gradually along the span and reaches its

extreme value near midspan or critical regions, reflecting the typical bending behavior of bridge structures under vertical loads.

In the longitudinal direction, a clear accumulation effect is observed. The maximum longitudinal displacement occurs at Node 72, with a value of approximately 20.892 mm. This displacement is mainly caused by thermal expansion and axial deformation of members, which becomes significant in long-span structures. Compared to the case without temperature effects, the increase in longitudinal displacement demonstrates the strong influence of temperature loading.

In addition, local displacement amplification is observed at nodes affected by support settlement, indicating that boundary condition variations directly influence the displacement field.

3.3.3 Rotation Response

The nodal rotation results show that the overall rotational response is relatively small. For example, the rotation about the z -axis at Node 2 is approximately -0.000377 rad, indicating a relatively high global stiffness of the structure. Combined with displacement results, it can be concluded that bending deformation dominates the structural response, while rotational effects remain secondary, which is consistent with the mechanical behavior of steel truss bridges.

3.3.4 Element Internal Forces

The analysis of local element forces (q_{local}) shows that axial force is the primary load transfer mechanism, consistent with the fundamental behavior of truss systems. Most truss members carry either tensile or compressive forces, while bending moments and shear forces remain relatively small.

The upper and lower chords form a typical tension–compression pair, while diagonal members transfer shear forces, establishing a clear load path. The maximum axial force occurs in Element 203, which uses section A3, with a value of approximately 3640 kN. The corresponding maximum stress is approximately 121.347 MPa, which is below the yield strength of structural steel, indicating that the design is within a reasonable engineering range.

Table 6. Critical Element Force and Stress

Element	Section	Axial Force (kN)	Stress (MPa)
203	A3	3640	121.347

3.3.5 Support Reactions

The support reactions show that the applied loads are primarily resisted at the supports, and the reaction distribution is generally consistent with the loading conditions. For supports without prescribed displacement changes, such as Nodes 1 and 39, the vertical reactions are approximately 2502.735 kN and 2497.265 kN, respectively, indicating a relatively symmetric load distribution.

For Nodes 38 and 76, where a vertical settlement of -50 mm is applied, the corresponding reactions are -11646.340 kN and 17166.342 kN. These values are significantly larger than those at other supports and exhibit clear asymmetry.

This phenomenon is not only caused by support settlement but also influenced by the non-uniform load distribution in the model. Since more nodes are located on the right side of the structure and each node carries a vertical load, the total load on that side is greater. As a result, the support at Node 76 carries a larger reaction force, while Node 38 exhibits a negative reaction due to settlement and local force redistribution.

These results demonstrate that support reactions are sensitive to both boundary conditions and load distribution, and that internal force paths are redistributed under combined effects, which is consistent with fundamental structural mechanics.

Table 7. Support Reactions

Node	Vertical Reaction (kN)	Condition
1	2502.735	Normal support
39	2497.265	Normal support
38	-11646.340	Settlement applied
76	17166.342	Settlement + load concentration

3.3.6 Special Effects Analysis

The model incorporates temperature effects, fabrication errors, and support settlement to simulate realistic engineering conditions. Temperature effects are introduced through a uniform temperature increase of $\Delta T = 10^{\circ}\text{C}$, resulting in global expansion and additional internal forces. Fabrication errors are simulated by introducing a length deviation of $\Delta L = -0.01$ m at the element level, altering the initial geometry and inducing additional forces. Support settlement is modeled by applying a prescribed vertical displacement of $u_y = -0.05$ m at selected nodes.

A comparative study is conducted between cases with and without temperature effects. The results show that vertical displacement is primarily governed by support settlement. Without temperature effects, the maximum vertical displacement is -50.277 mm, while with temperature effects it becomes -50.596 mm, indicating only minor differences.

In contrast, temperature has a significant impact on longitudinal displacement. The maximum longitudinal displacement increases from 6.488 mm (without temperature) to 20.892 mm (with temperature), demonstrating the accumulation of thermal expansion effects.

For transverse displacement, the maximum value decreases from 46.676 mm to 25.204 mm after introducing temperature effects. This indicates that thermal-induced axial forces enhance the global stiffness and modify the internal force distribution, thereby reducing lateral deformation.

Overall, temperature effects mainly influence longitudinal deformation, have limited impact on vertical displacement, and indirectly affect transverse displacement through stiffness redistribution.

Table 8. Key Displacement Results

Quantity	Node	Without Temp (mm)	With Temp (mm)
Max Vertical Displacement	34	-50.277	-50.596
Max Longitudinal Displacement	72	6.488	20.892
Max Transverse Displacement	–	46.676	25.204

3.3.7 Evaluation and Limitations

The results demonstrate that the structural model exhibits good engineering rationality in terms of deformation patterns, internal force distribution, and boundary response. The deformation profile is smooth, the maximum displacement occurs in expected regions, and the force distribution follows the fundamental behavior of truss systems. Support reactions satisfy global equilibrium, and additional effects such as temperature, fabrication errors, and settlement are properly captured.

However, several limitations exist in this study. Due to the lack of detailed real bridge data, the loading conditions are simplified. For example, self-weight and traffic loads are approximated using nodal forces and uniform loads, which may not accurately reflect real load distributions. As a result, the computed displacement magnitude is generally smaller than that expected in real structures.

In addition, the structural modeling involves idealizations in connection behavior. While truss and

frame elements are combined, connections are assumed to be either fully rigid or perfectly pinned. In reality, many connections exhibit semi-rigid behavior, which may affect stiffness distribution and force transfer.

Furthermore, temperature and fabrication effects are simplified as uniform changes, without considering gradients, material variability, or stochastic construction errors. These simplifications limit the ability of the model to capture more complex local responses.

4 Conclusion

The results demonstrate that the developed program can efficiently handle large-scale three-dimensional mixed structural systems and accurately compute displacement and internal force responses. The numerical results are consistent with fundamental engineering principles and successfully capture the effects of temperature, fabrication errors, and support settlement. Therefore, the program shows strong applicability and reliability for bridge structural analysis.

References

- [1] Xuhui He. *Structural Health Monitoring of the Nanjing Yangtze River Bridge and Its Key Technologies*. PhD thesis, Central South University, 2004.